Numerical Analysis Assignment 8

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Question 1

(a) **(i)** The interpolation polynomial is

$$p_2(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$

Its derivative is

$$\begin{aligned} p_2'(x) =& f[x_0, x_1] + f[x_0, x_1, x_2](2x - x_0 - x_1) \\ =& \frac{f(x_1) - f(x_0)}{x_1 - x_0} + \frac{\frac{f(x_2)-f(x_1)}{x_2-x_1} - \frac{f(x_1)-f(x_0)}{x_1-x_0}}{x_2 - x_0}(2x - x_1 - x_0) \end{aligned}$$

Let $x_0 = x - h$, $x_1 = x$, $x_2 = x + 3h$, then

$$\begin{aligned} p_2'(x) =& \frac{f(x) - f(x-h)}{h} + \frac{\frac{f(x+3h)-f(x)}{3h} - \frac{f(x)-f(x-h)}{h}}{4h}h \\ =& \frac{8f(x) - 9f(x-h) + f(x+3h)}{9h} \end{aligned}$$

Then the estimate of $f'(x)$ is

$$D_h^{(1)} = \frac{8f(x) - 9f(x-h) + f(x+3h)}{9h}.$$

(ii) The Taylor series of $f(x)$, $f(x-h)$ and $f(x+3h)$ is

$$\begin{aligned} f(x) =& f(x) \\ f(x-h) =& f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(c_1) \\ f(x+3h) =& f(x) + 3hf'(x) + \frac{8h^2}{2}f''(x) + \frac{27h^3}{6}f'''(c_2) \end{aligned}$$

Let $D_h^{(1)} = Af(x) + Bf(x-h) + Cf(x+3h)$, then

$$\begin{cases} A + B + C = 0 \\ -Bh + 3Ch = 1 \\ Bh^2/2 + 9h^2C/2 = 0 \end{cases}$$

The solution of the system is $A = 3/2h$, $B = -3/4h$ and $C = 1/12h$ So, the estimate of $f'(x)$ is

$$D_h^{(1)} = \frac{8f(x) - 9f(x-h) + f(x+3h)}{12h}.$$

(b) There are infinite many first-order formula for approximating the first derivative $f'(x)$ that only use $f(x)$, $f(x-h), f(x+3h)$. For example

$$D_h^{(1)} = \frac{-3f(x) + f(x-h) + f(x+3h)}{2h}.$$

Question 2

The Taylor series of $f(x)$, $f(x-h)$ and $f(x+2h)$ is

$$\begin{aligned} f(x) =& f(x) \\ f(x+h) =& f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(c_1) \\ f(x-2h) =& f(x) - 2hf'(x) + \frac{2h^2}{2}f''(x) - \frac{8h^3}{6}f'''(c_2) \end{aligned}$$

Then

$$E = f'(x) - \frac{4f(x+h) - 3f(x) - f(x+2h)}{6h} = \frac{h^2}{6}\left(\frac{2f'''(c_1) + 4f'''(c_2)}{3}\right) = \frac{h^2}{3}.$$

The formula is second order.

Question 3

By integration by part,

$$\int_0^\pi x\cos x \mathrm{d}x = (x\sin x + \cos x)\big|_0^\pi = -2.$$

The approximation and errors is shown in the table below

	In (a)		In (b)		In (c)	
m	Integral	Error	Integral	Error	Integral	Error
1	0.0000	2.0000	-4.9348	2.9348	-1.6449	0.3551
2	-1.7447	0.2553	-2.4674	0.4674	-1.9856	0.0144
3	-1.8994	0.1006	-2.1932	0.1932	-1.9974	0.0026